6/12/25

Graph Algorithms

Warshall's Algorithm

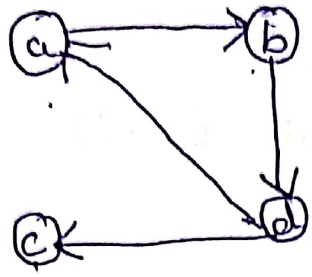
* It is also known as. Floyd - washall Algorithm.
* Dynamic Programming approach.
* Find the Shortest paths between all pairs of vertices in a weighted, directed graph.
* Handles both positive and negative edge weights but does not work with negative cycles.
* Computes the transitive closure of a relation.

Transitive closure

* It is the process of creating a new ~~relatio~~ relation that includes all original relations plus any additional ones required to make the entire relation transitive.

*

Eg:



$$A = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

transitive closure

$$\begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \end{array}$$

Algorithm

A Warshall$(A[1 \ldots n, 1 \ldots n])$

//Implements warshall's algorithm for computing transitive closure.

// Input: The adjacency matrix A of a diagraph with n vertices

//output: The transitive closure of the diagraph.

$R^{(0)} \leftarrow A$

for $k \leftarrow 1$ to n do

for $i \leftarrow 1$ to n do

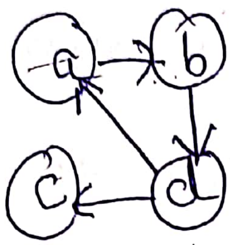
for $j \leftarrow 1$ to n do

$R^{(k)}[i,j] \leftarrow R^{(k-1)}[i,j]$ or $(R^{(k-1)}[i,k]$ and $R^{(k-1)}[k,j])$

return $R^{(n)}$.

Eg:

Transitive closure ^Path matrix of a directed graph or all paths in a directed graph using adjacency matrix.



} Finding every path of vertices

1) Generate adjacency matrix

$$R^{(0)} = \begin{array}{c|cccc} & a & b & c & d \\ \hline a & 0 & 1 & 0 & 0 \\ b & 0 & 0 & 0 & 1 \\ c & 0 & 0 & 0 & 0 \\ d & 1 & 0 & 1 & 1 \end{array}$$

2) Consider path through vertex a

$$R(0) = \begin{array}{c|cccc} & a & b & c & d \\ \hline a & 0 & 1 & 0 & 0 \\ b & 0 & 0 & 0 & 1 \\ c & 0 & 0 & 0 & 0 \\ d & 1 & 0 & 1 & 0 \end{array}$$

$$R^1 = \begin{array}{c|cccc} & a & b & c & d \\ \hline a & 0 & 1 & 0 & 0 \\ b & 0 & 0 & 0 & 1 \\ c & 0 & 0 & 0 & 0 \\ d & 1 & 1 & 1 & 0 \end{array}$$

Consider graph in between vertex a

$b \to b = b \to a \; \& \; a \to b$
$= 0 \; \& \; 1 = 0$

$b \to c = b \to a \; \& \; a \to c$
$= 0 \; \& \; 0 = 0$

$c \to b = c \to a \;\&\; a \to b = 0 \;\&\; 1 = 0$

$c \to c = c \to a \;\&\; a \to c = 0 \;\&\; 0 = 0$

$c \to d = c \to a \;\&\; a \to d = 0 \;\&\; 0 = 0$

$d \to b = d \to a \;\&\; a \to b = 1 \;\&\; 1 = 1$

$d \to d = d \to a \;\&\; a \to d = 1 \;\&\; 0 = 0$

$$R^1 = \begin{array}{c|cccc} & a & b & c & d \\ \hline a & 0 & 1 & 0 & 0 \\ b & 0 & 0 & 0 & 1 \\ c & 0 & 0 & 0 & 0 \\ d & 1 & 1 & 1 & 0 \end{array}$$

Path through vertex (b)

$$R^2 = \begin{array}{c|cccc} & a & b & c & d \\ \hline a & 0 & 1 & 0 & 1 \\ b & 0 & 0 & 0 & 1 \\ c & 0 & 0 & 0 & 0 \\ d & 1 & 1 & 1 & 1 \end{array}$$

$a \to a = a \to b \;\&\; b \to a = 1 \;\&\; 0 = 0$

$a \to c = a \to b \;\&\; b \to c = 1 \;\&\; 0 = 0$

$a \to d = a \to b \;\&\; b \to b = 1 \;\&\; 1 = 1$

$c \to a = c \to b \;\&\; b \to a = 0 \;\&\; 0 = 0$

$c \to c = c \to b \;\&\; b \to c = 0 \;\&\; 0 = 0$

$c \to d = c \to b \;\&\; b \to d = 0 \;\&\; 1 = 0$

$d \to d = d \to b$
$\&\; b \to d$
$1 \;\&\; 1 = 1$

Path vertex c

$$R^{(3)} = \begin{array}{c|cccc} & a & b & c & d \\ \hline a & 0 & 1 & 0 & 1 \\ b & 0 & 0 & 0 & 1 \\ c & 0 & 0 & 0 & 0 \\ d & 1 & 1 & 1 & 1 \end{array}$$

$a \to a = a \to c$ & $c \to a = 0$ & $0 = 0$

$b \to a = b \to c$ & $c \to a = 0$ & $0 = 0$

$b \to b = b \to c$ & $c \to b = 0$ & $0 = 0$

Path Vertex d

$$R^{4} = \begin{array}{c|cccc} & a & b & c & d \\ \hline a & 1 & 1 & 1 & 1 \\ b & 1 & 1 & 1 & 1 \\ c & 0 & 0 & 0 & 0 \\ d & 1 & 1 & 1 & 1 \end{array}$$

$a \to a = a \to d$ & $d \to a = 1$ & $1 = 1$

$a \to c = a \to d$ & $d \to c = 1$ & $1 = 1$

$b \to a = b \to d$ & $d \to a = 1$ & $1 = 1$

$b \to b = b \to d$ & $d \to b = 1$ & $1 = 1$

$b \to c = b \to d$ & $d \to c = 1$ & $1 = 1$

$c \to a = c \to d$ & $d \to a = 0$ & $1 = 0$

$c \to b = c \to d$ & $d \to b = 0$ & $1 = 0$

$c \to c = c \to d$ & $d \to c = 0$ & $1 = 0$

Final answer → Transitive closure

$$R^{(4)} = \begin{array}{c|cccc} & a & b & c & d \\ \hline a & 1 & 1 & 1 & 1 \\ b & 1 & 1 & 1 & 1 \\ c & 0 & 0 & 0 & 0 \\ d & 1 & 1 & 1 & 1 \end{array}$$

→ Transitive closure | Path matrix

Reach all the paths.